

Prerequisites of DL

Revision of Important concepts of Machine Learning

The diagram consists of three concentric ellipses. The outermost ellipse is light blue and labeled 'Artificial Intelligence'. Inside it is a medium blue ellipse labeled 'Machine Learning'. The innermost ellipse is dark blue and labeled 'Deep Learning'. Each ellipse contains a definition of that term.

Artificial Intelligence

Machine Learning

Deep Learning

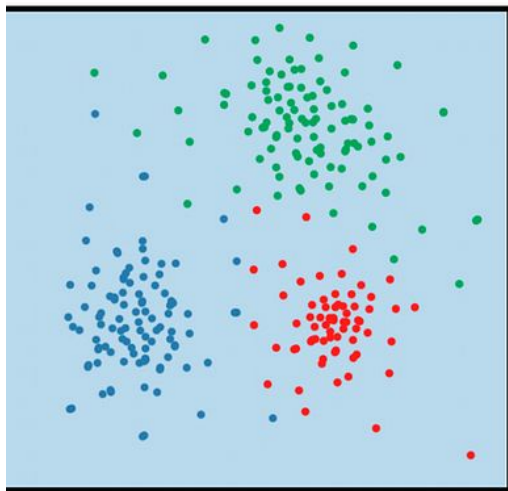
The subset of machine learning composed of algorithms that permit software to train itself to perform tasks, like speech and image recognition, by exposing multilayered neural networks to vast amounts of data.

A subset of AI that includes abstruse statistical techniques that enable machines to improve at tasks with experience. The category includes deep learning

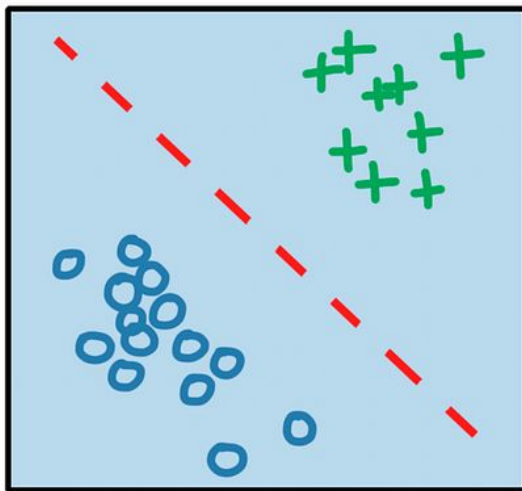
Any technique that enables computers to mimic human intelligence, using logic, if-then rules, decision trees, and machine learning (including deep learning)

machine learning

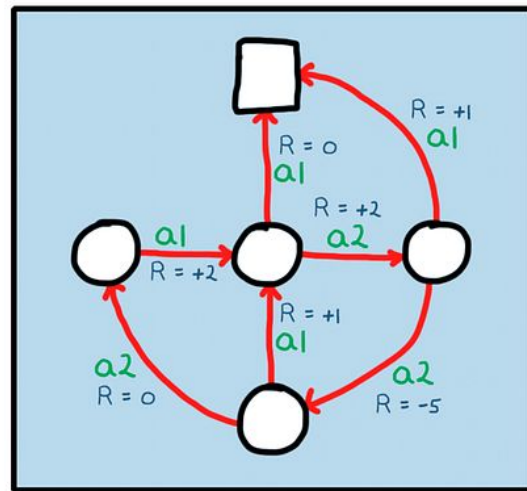
unsupervised
learning



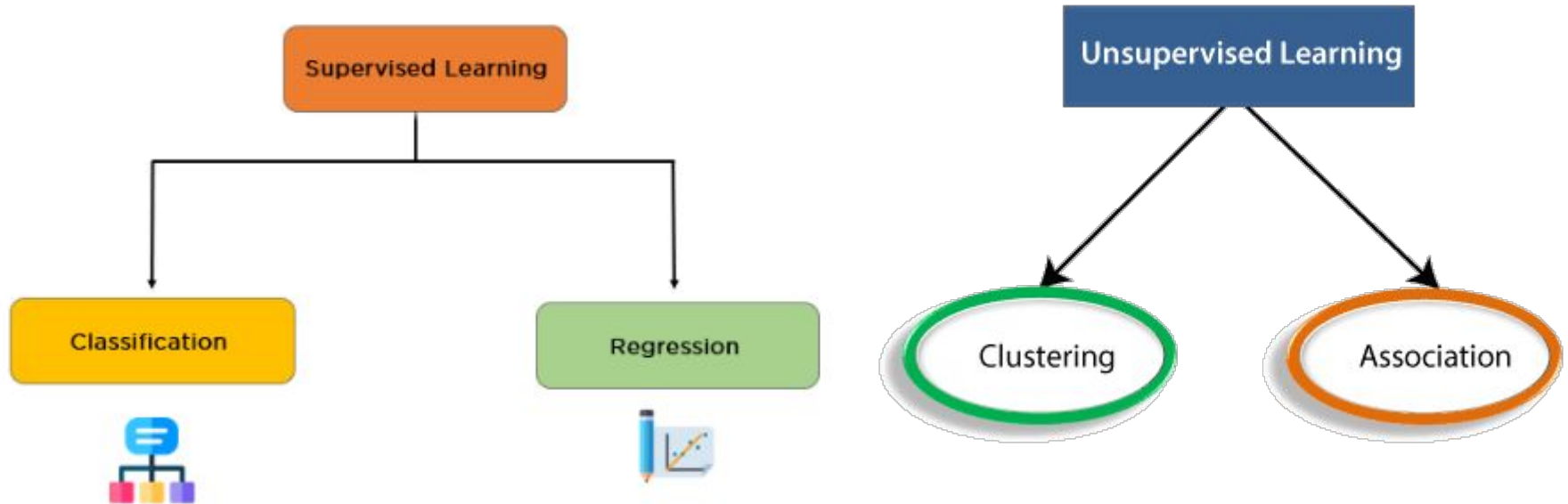
supervised
learning



reinforcement
learning



Types of Supervised and Unsupervised Machine learning



Linear Regression

Linear Regression

Key Idea

The target y can be expressed as a **linear combination** of input features X plus an error term:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n + \epsilon$$

- y : Dependent variable (output we want to predict)
- $x_1, x_2, \dots, x_n$: Independent variables (inputs/features)
- β_0 : Intercept (value of y when all x 's are 0)
- $\beta_1, \dots, \beta_n$: Coefficients (weights showing how much each x affects y)
- ϵ : Error term (difference between predicted and actual values)

OLS (Ordinary Least Squares)

Steps

1. Collect dataset with inputs (X) and outputs (y).
2. Assume a linear model to predict output.
3. Define error as squared difference between predictions and actual values.
4. Minimize this error function.
5. Solve using derivative = 0.
6. Get solution using matrix inverse.

Mathematics

Dataset: $X \in \mathbb{R}^{n \times p}, y \in \mathbb{R}^n$

$$y \approx X\beta + \epsilon$$

Loss: $L(\beta) = |y - X\beta|^2$

$$\hat{\beta} = \arg \min_{\beta} L(\beta)$$

$$X^T X \hat{\beta} = X^T y$$

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

Gradient Descent (GD / SGD)

Steps in Easy English

1. Initialize weights randomly.
2. Predict outputs using current weights.
3. Compute error and gradient of loss.
4. Update weights in opposite direction of gradient.
5. Repeat until convergence (error small).
6. Return final weights.

Mathematical Steps

$$\beta^{(0)} \sim \text{random}$$

$$\hat{y}^{(t)} = X\beta^{(t)}$$

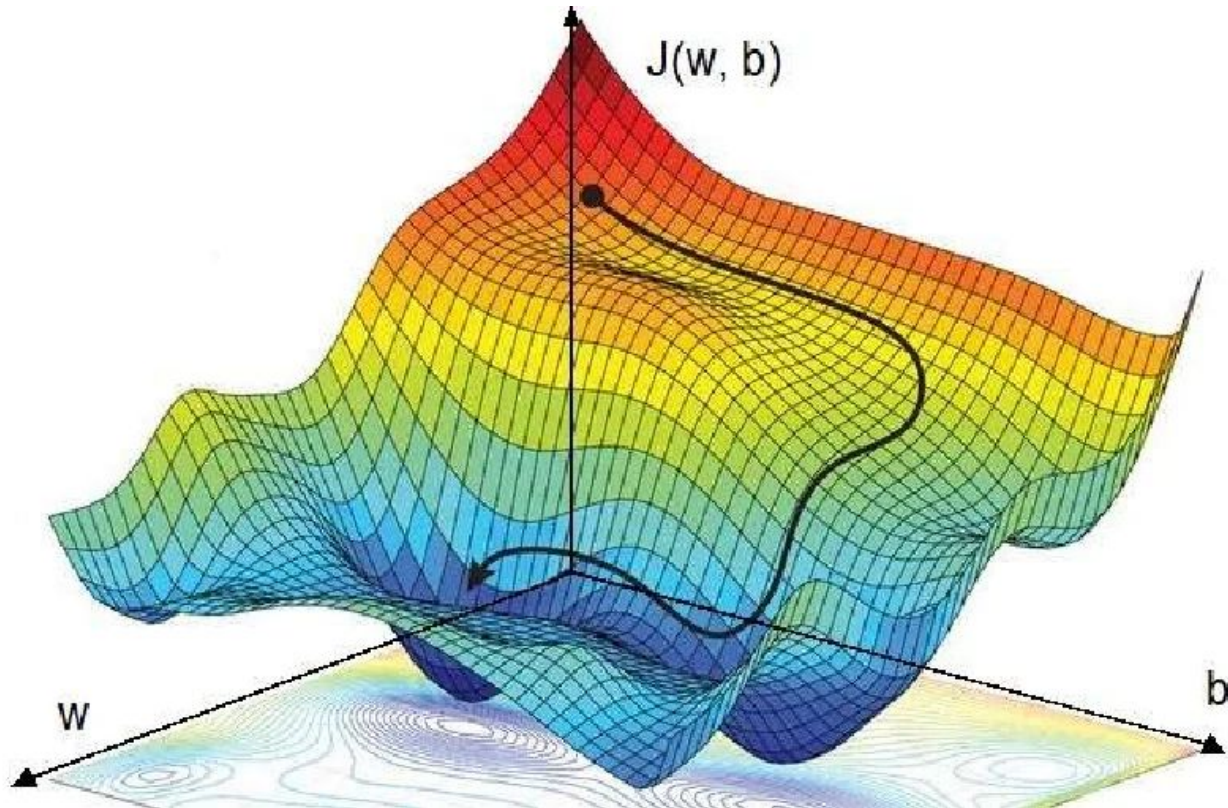
$$\nabla_{\beta} L(\beta) = -2X^T(y - X\beta)$$

$$\beta^{(t+1)} = \beta^{(t)} - \eta \nabla_{\beta} L(\beta^{(t)})$$

Stop when $|\nabla_{\beta} L(\beta)| < \epsilon$

$$\hat{\beta} = \beta^{(T)}$$

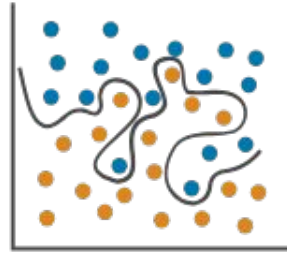
Gradient Descent visualization



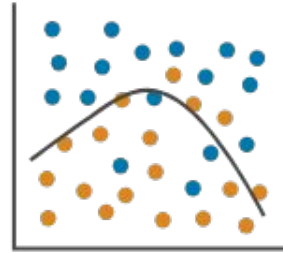
Overfitting and Underfitting

Classification

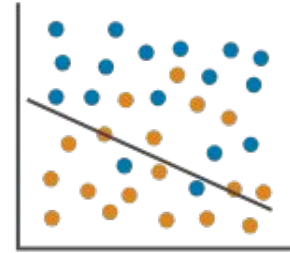
Overfitting



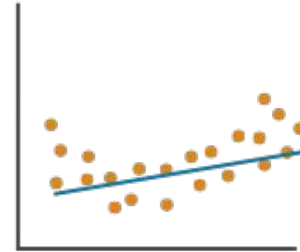
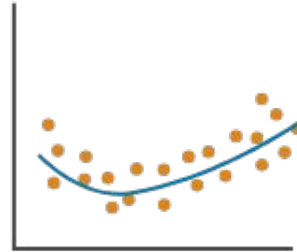
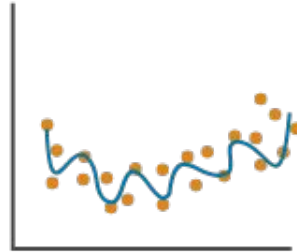
Right Fit



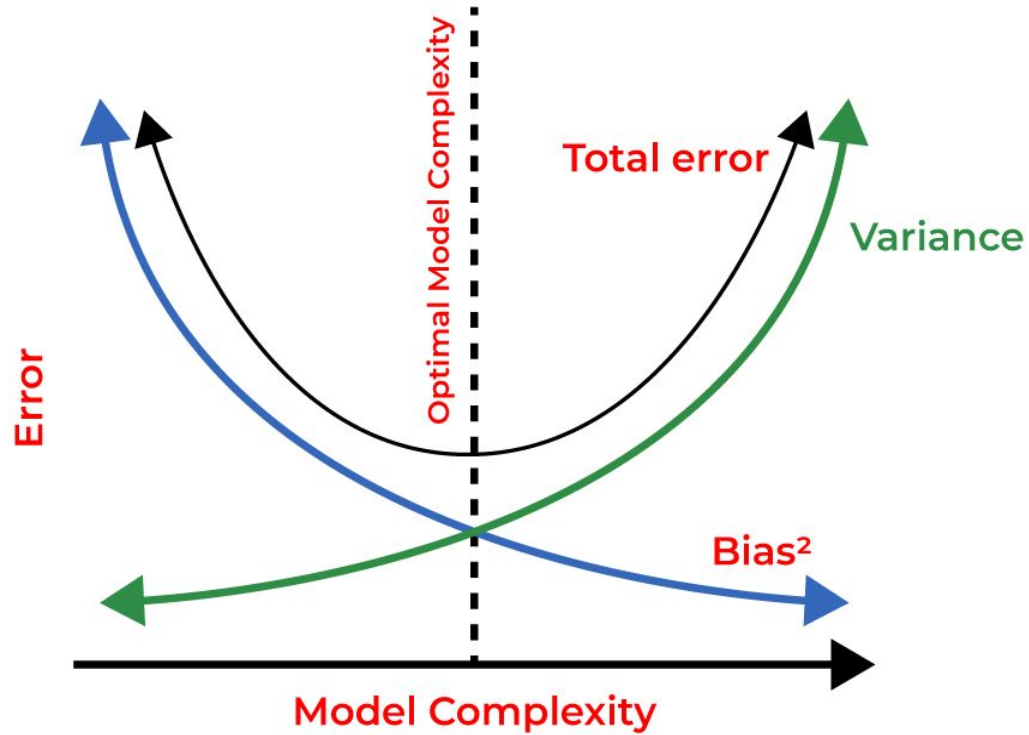
Underfitting



Regression

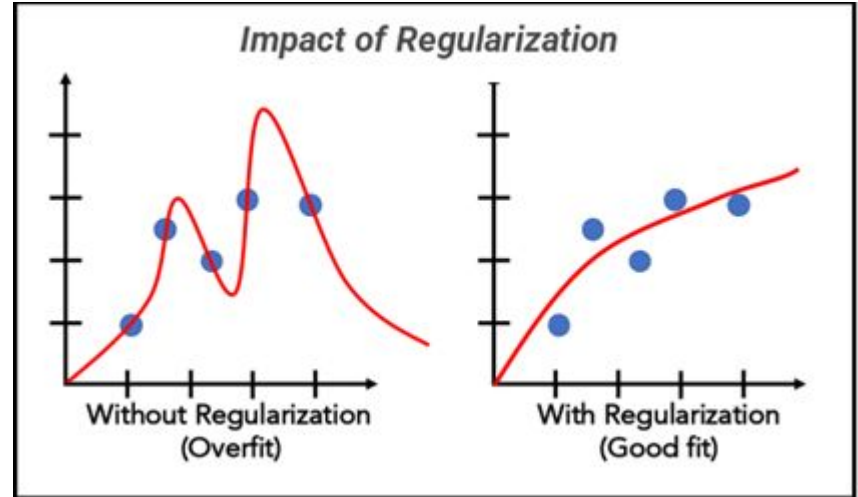


Bias and Variance



Why Regularization?

1. Overfitting = model memorizes training data
 - a. High variance $\rightarrow$ poor generalization
 - b. Solution: control model complexity
2. Key Idea:
 - a. Add a penalty to discourage overly complex models
3. Connection (very important for flow):
 - a. Bias–Variance tradeoff (previous slide)
 - b. Regularization = variance reduction tool



Regularization in Objective Function

$$J(W) = \mathcal{L}(W) + \lambda \cdot \Omega(W)$$

- Small $\lambda \rightarrow$ focus on fitting data
- Large $\lambda \rightarrow$ focus on simplicity
- $\mathcal{L}(W)$: data loss
- $\Omega(W)$: regularization term
- λ : controls strength

L2 Regularization (Ridge → Weight Decay)

1. Key Points:

- a. Penalizes large weights
- b. Smooth shrinkage
- c. No weights become exactly zero

2. Update Insight:

- a. Weights shrink every step → “decay”

$$J(W) = \mathcal{L}(W) + \lambda \sum ||W||^2$$

3. Effect:

- a. Stable learning
- b. Better generalization

L1 Regularization (Lasso → Sparsity)

Key Points:

Constant push toward zero

Many weights become exactly zero

$$J(W) = \mathcal{L}(W) + \lambda ||W||_1$$

Effect:

Feature selection

Sparse models

Model Evaluation & Optimization

Regression Metrics

1. Error-Based Metrics

Mean Absolute Error (MAE)

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

- Measures the average absolute difference between predicted and actual values.
- All errors are treated equally.
- Lower values indicate better performance.

Mean Squared Error (MSE)

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

- Measures the average squared difference between predictions and actuals.
- Penalizes larger errors more heavily.
- Useful when large deviations are undesirable.

Root Mean Squared Error (RMSE)

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

- Square root of MSE; in the same units as the target variable.
- Commonly used for interpretability and overall model accuracy.

R^2 Score (Coefficient of Determination)

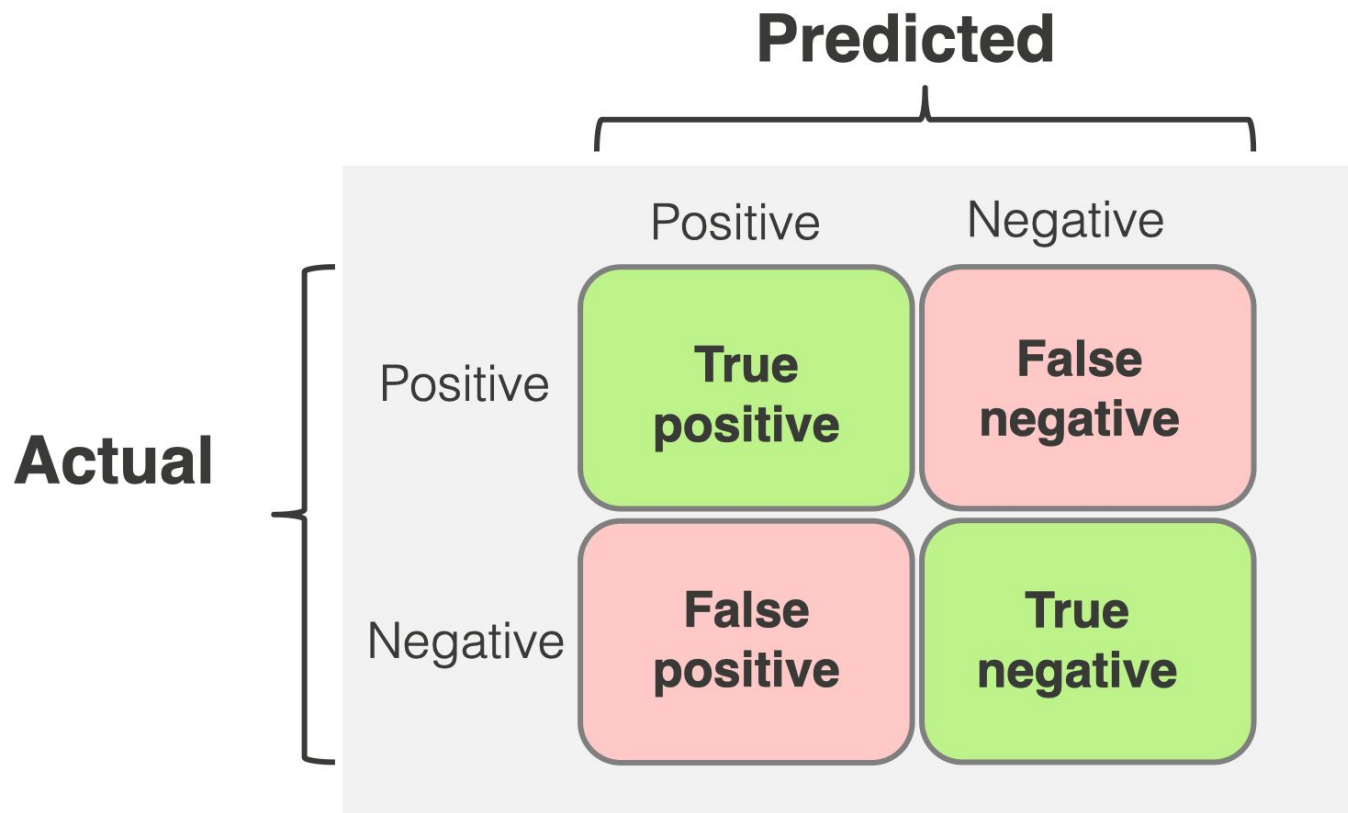
$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

- Represents the proportion of variance explained by the model.
- Range: $-\infty, 1]$
- $R^2 = 1$: Perfect prediction
- $R^2 = 0$: Model as good as mean prediction
- $R^2 < 0$: Worse than mean prediction

Adjusted R^2

$$R_{adj}^2 = 1 - (1 - R^2) \frac{n - 1}{n - p - 1}$$

- Adjusts R^2 for the number of predictors p .
- Useful when comparing models with different feature counts.



Classification Metrics

1. Accuracy

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

- Measures the overall correctness of the model.
- Works well when classes are balanced.
- Can be misleading for imbalanced datasets.

Example:

If 90 out of 100 predictions are correct → Accuracy = 90%.

2. Precision

$$Precision = \frac{TP}{TP + FP}$$

- Fraction of correctly predicted positive observations to all predicted positives.
- High precision means fewer false positives.

Use case: When the cost of a false positive is high (e.g., spam detection).

3. Recall (Sensitivity or True Positive Rate)

$$Recall = \frac{TP}{TP + FN}$$

- Fraction of correctly predicted positives to all actual positives.
- High recall means fewer false negatives.

Use case: When missing a positive case is costly (e.g., disease detection).

4. F1-Score

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

- Harmonic mean of precision and recall.
- Balances the trade-off between precision and recall.
- Useful when data is imbalanced.

5. Specificity (True Negative Rate)

$$\textit{Specificity} = \frac{TN}{TN + FP}$$

- Measures how well the model identifies negative cases.
- Complements Recall for balanced evaluation.

6. Confusion Matrix

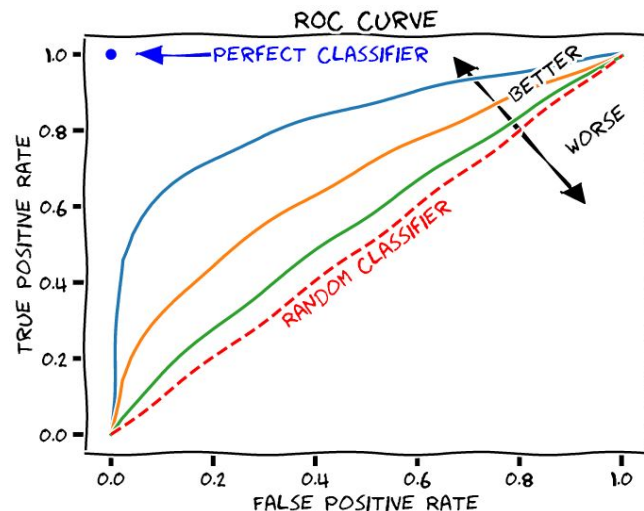
	Predicted Positive	Predicted Negative
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

- Summarizes correct and incorrect predictions.
- Useful for calculating all other metrics.

7. ROC Curve and AUC (Area Under Curve)

- **ROC Curve:** Plots True Positive Rate vs. False Positive Rate.
- **AUC:** Measures the entire two-dimensional area under the ROC curve.
- **Interpretation:**
 - $AUC = 1 \rightarrow$ Perfect model
 - $AUC = 0.5 \rightarrow$ Random model

Use case: Evaluating classifier performance across all thresholds.



8. Log Loss (Cross-Entropy Loss)

$$\text{LogLoss} = -\frac{1}{n} \sum_{i=1}^n [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)]$$

- Penalizes false predictions with confidence.
- Lower value indicates better calibrated probabilities.
- Used in probabilistic classifiers (e.g., logistic regression).

One-slide Summary

Problem Type	Default DL Loss	Evaluation Metric
Regression	MSE / MAE	RMSE, R^2
Binary Classification	Binary Cross-Entropy	Precision, Recall, F1, AUC
Multi-class	Cross-Entropy	Accuracy, F1

ML vs DL Concept Mapping

ML Concept	In Machine Learning	In Deep Learning
Model	Linear / simple models	Deep neural networks
Parameters	Few (10–1000)	Millions–billions
Loss Function	MSE, Logistic loss	Cross-entropy, custom losses
Gradient Descent	Used for optimization	Same, but with SGD, Adam, etc.
Regularization	L1, L2	Weight decay + Dropout + Augmentation
Overfitting	Moderate risk	Extremely high risk

ML vs DL Concept Mapping

ML Concept	In Machine Learning	In Deep Learning
Feature Engineering	Manual	Learned automatically
Optimization	Convex (often)	Non-convex
Data Requirement	Small–medium	Very large datasets
Generalization	Controlled by model choice	Controlled by regularization
Bias-Variance	Simple tradeoff	Complex, high-dimensional tradeoff
Evaluation	Metrics (MSE, Accuracy)	Same + task-specific metrics